Security Focus: Fork Bombs

Now that you understand how processes work, we can look at an example of how an understanding of them can help protect computer systems.



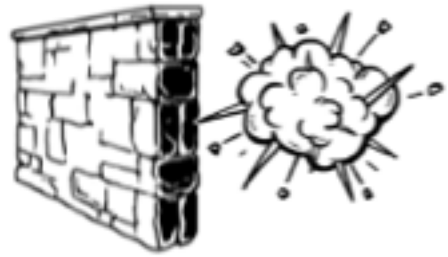
An attack called a fork bomb, otherwise known as a rabbit virus or wabbit, targets a system by launching many new processes until no more new processes can be created.



This Denial of Service (DoS) attack usually crashes the system completely.
"Denial of Service attack? Wait - what's that?" you ask.

Well, recall the "Availability" classification from the information security CIA triad. A Denial of Service attack prevents users from being able to use systems when they need them and, therefore, is classified as an attack on availability.

A fork bomb attack creates a new process, which creates another new process, which, in turn, creates yet another new process and so on and so forth.   Well, you get the idea. Eventually, the process table fills until no more new processes can be created.

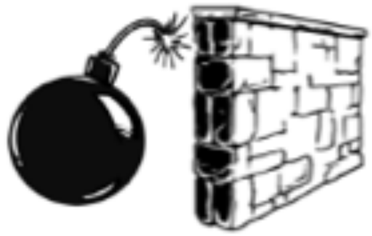


The good news is that system administrators can easily prevent this attack by setting limits on how many processes a user can initiate.

#!/bin/bash

ulimit -u 40

In Linux, this BASH command is ulimit with the -u flag. So to limit a user to 40 processes, I would type ulimit -u 40.



Prevention of the fork bomb attack is just one example of how an understanding of processes is important for future IT professionals.